

AREENA MAHEK

AI Software Engineer | Full-Stack Engineer

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SUMMARY

AI-Driven Software Engineer with 2+ years of experience building cloud-integrated, scalable applications. Strong in Next.js, React.js, Python, and Golang, with hands-on expertise integrating n8n workflows, vector databases, and LLMs (OpenAI, Claude) to deliver intelligent, user-centric products.

RELEVANT EXPERIENCE

Oxmaint Inc. | Software Engineer (Remote) | Sunnyvale, CA

Jan 2026 – Present

- Designed and enhanced core product features and AI-driven automation workflows; delivered customized solution design for enterprise customers.
- Implemented third-party integrations with QuickBooks and SAP to support enterprise workflows.
- Applied data analysis to improve system performance, scalability, quality assurance and reliability.
- Participated in Agile development including sprint planning, execution and reviews.

Oxmaint Inc. | Software Engineer Intern (Remote) | Sunnyvale, CA

Sept 2025 – Dec 2025

- Contributed to the development of the Enterprise Asset Management (EAM) web portal using React, Next.js, and TypeScript (frontend) and Golang (backend).
- Built and maintained AI/automation orchestration workflows using n8n.
- Designed and implemented full-stack features, including API integrations and UI components, ensuring scalable and maintainable code.
- Integrated LLMs (GPT-OSS, Anthropic Claude) to enhance product features and reduce customer onboarding by 40% through automated, customer-specific data generation.

DePaul University | Research Assistant | Chicago, IL

Scalable Data Management Service as a Microservice – ByteBridge

Jan 2025 – June 2025

- Designed a modular subsystem with Django + REST APIs (backend) and HTML/CSS/JavaScript (frontend), improving research data accessibility and visualization by 70%.
- Ensured reliability through Agile practices, Git-based version control, code reviews, and unit testing while collaborating with cross-functional teams to resolve development challenges and support seamless platform performance.

Interactive Out-of-Distribution (OoD) Data Visualization Dashboard

Aug 2024 – Dec 2024

- Developed an interactive Visualizer Dashboard using Python, Trame and Vuetify (Vue.js) to support anomaly detection in high-dimensional Lung Nodule image data (LIDC Data Collection) from radiologists.
- Implemented threshold-based filtering to categorize images by OoD scores, improving interpretability and aiding model debugging

DePaul University | Graduate Assistant-Tutoring Mentor | Chicago, IL

Sept 2023 – June 2025

- Mentored and supported over 200 undergraduate and graduate students in Python, Java, SQL, Django, Web Development, Tableau, Power BI, Analysis of Algorithms, Distributed Systems and Data Structures while achieving a 95% student performance improvement rate.
- Assisted in debugging, optimizing code, and guiding development projects in distributed systems and full-stack apps.

SKILLS

- Frontend:** JavaScript (ES6+), TypeScript, React.js, Next.js, Vue.js, Vuetify, Tailwind CSS, HTML5, Bootstrap
- Backend & APIs:** Golang, Python (Django, Flask), REST APIs, Trame
- AI & Machine Learning:** OpenAI, Anthropic Claude, Retrieval-Augmented Generation (RAG), Vector Databases (MongoDB, Milvus), NLP (NLTK, Hugging Face), ML Fundamentals (Classification, Clustering)
- Enterprise Integrations:** SAP, QuickBooks
- Data Visualization & Analytics:** Python, R, SQL (Oracle, MySQL, SQLite), Pandas, NumPy, SciPy, Scikit-learn, Tableau, Power BI, Matplotlib, Plotly, Seaborn, VTK
- Cloud & Big Data:** AWS (S3, EMR, EC2, Athena), Azure, Hadoop, Apache Hive, Google BigQuery
- Engineering & DevOps:** Docker, Git, Linux, JWT, OAuth2, Data Structures & Algorithms, Unit Testing, Web Scraping (BeautifulSoup)
- Soft Skills:** Problem-Solving, Communication, Collaboration, Adaptability, Time Management

OTHER PROJECTS & ANALYTICS EXPERIENCE

Cloud-Based Sentiment Analysis on Big Data using AWS Services

Nov 2024

- Performed data acquisition and preprocessing on 10 million Amazon product reviews and stored it in AWS S3.
- Utilized AWS EMR to build a scalable data pipeline for modeling, sentiment analysis, and visualizing insights derived from the dataset.

Online Customer Behavior Prediction with Data Analytics and Machine Learning

Jan 2024

- Developed an automated pipeline for online customer behavior prediction, including data collection, exploratory data analysis and feature engineering; applied random sampling for class balance and trained models using ensemble methods (Stacking Classifier).
- Achieved 85% accuracy and evaluated performance using F1-score, ROC curve and AUC, optimized results through metric-driven tuning.

EDUCATION

DePaul University | Chicago, IL, USA

M.S. in Computer Science with Data Science | GPA: 3.96/4.0

June 2025

Stanley College of Engineering and Technology for Women | Hyderabad, India

MTech. in Computer Science | GPA: 3.6/4.0

Dec 2022

Muffakhham Jah College of Engineering and Technology | Hyderabad, India

B.E. in Information Technology | GPA: 3.75/4.0

Sept 2020